

Tharun Vaibhav S S

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Profile Summary

MCA student specializing in Artificial Intelligence, Generative AI, LLM-powered applications, Python, and Data & Analytics, with hands-on experience developing AI and IoT solutions and full-stack applications. Experienced in integrating OpenAI GPT-5.5 API into AI-powered industrial diagnostics, developing predictive models and automated reporting solutions. Strong foundation in Python, FastAPI, SQL, PostgreSQL, data analysis, application development, debugging, testing, and performance optimization. Demonstrated ability to research and implement AI-driven solutions, rapidly prototype ideas, collaborate with teams, and solve real-world technical problems.

Education

Master of Computer Application (MCA) - 78%*	May 2027
PSG College of Arts & Science	
B.Sc Information Technology - 75.9%	May 2025
PSG College of Arts & Science	

Internship

Software Development Intern Dyzen Consultants Duration: 10 Months	2026
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- Developed and deployed 4+ web applications, contributing to application development, testing, maintenance, and reliable software delivery.
- Performed debugging, testing, issue identification, and performance optimization to improve application reliability and user experience.
- Collaborated with cross-functional teams to understand requirements, investigate technical issues, and implement practical solutions.
- Monitored application performance and responsiveness to identify potential areas for improvement.
- Used analytics and user feedback to identify application and user-experience issues, contributing to a 25–30% reduction in bounce rates.
- Worked with WordPress, Figma, and modern UI/UX practices while delivering solutions based on client requirements

Skill Set

- Programming: Python, Java, Next.js
- Frameworks: FastAPI, Django
- Web Technologies: HTML, CSS, WordPress
- Deployment: Vercel, Hostinger, Render, GitHub
- AI Tools: Anti-Gravity, Bolt.new, Lovable.ai
- Databases: MySQL, Oracle SQL, MongoDB, PostgreSQL
- Data Analytics tools: Power BI, Tableau, Excel
- Soft Skills: Adaptive Learner, Team Facilitator, Design Thinker, Self-Driven

Projects

Enervision - AI Energy Audit System 2026	PSGCAS
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Link: github.com/tharunvaibhavss/enervision

- Developed an AI and IoT-based energy monitoring system for real-time energy consumption analysis, anomaly detection, and intelligent monitoring.

- Implemented predictive models to forecast energy usage and support data-driven troubleshooting, optimization, and decision-making.
- Applied data analysis and intelligent monitoring techniques to improve energy efficiency by 20–25
- Designed visualization and reporting capabilities to communicate system performance, energy insights, and sustainability findings.
- Developed the solution as an AI/IoT proof-of-concept focused on real-world energy analytics and optimization.

AI-Powered Industrial Machine Diagnostic System 2026

PSGCAS

Link: github.com/tharunvaibhavss/cat

- Developed a full-stack AI-powered industrial machine diagnostic system for equipment monitoring, predictive diagnostics, maintenance support, work order management, and automated reporting.
- Integrated the OpenAI GPT-5.5 API and LLM capabilities to provide AI-assisted machine diagnostics and maintenance recommendations.
- Built AI-driven workflows covering telemetry monitoring, machine management, manual inspections, vision inspections, and automated diagnostic reporting.
- Developed backend services using Python, FastAPI, PostgreSQL, SQLAlchemy, and JWT, supporting scalable application workflows and structured system access.
- Implemented role-based authentication and application workflows for controlled access to machine and diagnostic information.

Client Website Projects 2026

Dyzen Consultants

Link: github.com/tharunvaibhavss/websites

- Developed and deployed 4+ responsive business applications/websites, contributing to application development, testing, maintenance, and reliable software delivery.
- Performed debugging, testing, performance optimization, and issue identification to improve application reliability and user experience.
- Collaborated with cross-functional teams to understand requirements, investigate technical issues, and implement solutions based on client needs.
- Used WordPress, Figma, modern UI/UX practices, and web technologies to develop and maintain client solutions.principles.

Certifications

- Python for Data Science – NPTEL
- Data Science 101 – IBM
- Linux Shell Programming – PSG College of Arts & Science
- Power BI for Beginners – Simplilearn
- Network Security – Great Learning
- AI and Business Strategy: Case Studies – LinkedIn

Co-Curricular Activities

- Best Performance — IoT-Based Technology Innovative Expo 2026: Recognized for innovation and technical excellence in IoT solutions.
- Best Minimum Viable Product (MVP) — Hack-Arti-Thon 2.0, 2026: Recognized for rapid prototyping, problem solving, and development of an innovative solution during a 24-hour national-level hackathon.
- Smart India Hackathon 2023: Collaborated on solutions for Student Dropout Analysis and AICTE innovation challenges.
- Socio-Tech Hackathon 2024: Developed solutions addressing real-world anti-drug challenges.
- Hackverse 2026 Mini Hackathon: Participated as a student coordinator supporting ideation and innovation-driven development.